

ANEESHA AVASTHI

Graduate Research Assistant — Cleveland, Ohio, USA

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OBJECTIVE

Passionate researcher working to advance detector technologies used in particle physics experiments. In addition to building detectors, I work on simulating particle interactions in gas detectors. I also enthusiastic about the applications of physics technologies to further developments in medical science, environmental protection, and outreach.

EDUCATIONAL BACKGROUND

Case Western Reserve University, Ohio, USA

Ph.D. Physics, Department of Physics

Started Aug. 2020

GPA: 4.0/4.0

State University of New York at Buffalo, New York, USA

Ph.D. Physics (transferred out), Department of Physics

Aug. 2019 – May 2020

GPA: 3.8/4.0

Indian Institute of Technology Guwahati, INDIA

M.Sc. Physics, Department of Physics

Jul. 2017 – May 2019

GPA: 8.3/10.0

Sri Guru Tegh Bahadur Khalsa College, University of Delhi, INDIA

B.Sc. Physics (Honours), Department of Physics & Astrophysics

Jul. 2014 – May 2017

Percentage: 91.36%

Rani Laxmi Bai Memorial School, INDIA

Senior Secondary School

Apr. 2013 – Mar 2014

Percentage: 95.60%

Rani Laxmi Bai Memorial School, INDIA

High School

Apr. 2011 – Mar 2012

GPA: 10.0/10.0

RESEARCH EXPERIENCE

Case Western Reserve University (CWRU)

Graduate Research Assistant, Advisor: Dr. Benjamin Monreal

Aug. 2020 – Present

Department of Physics

- Developed a proportional detector to characterize Se-based compounds to demonstrate their feasibility as effective drift gases in TPCs.
- Working on simulating particle tracks in time projection chambers to reconstruct interactions between dark matter and noble gas nuclei.

Indian Institute of Technology Guwahati (IITG)

Master's thesis, Advisor: Dr. Bosanta Ranjan Boruah

Dec. 2018 - Apr. 2018

Department of Physics

- Project Title: Contiguous sub-wavelength manipulation of light focused by a lens
- Simulated diffraction patterns using MATLAB and verified the results experimentally.
- Implemented phase extraction of a modified wavefront to reconstruct the original masking of damaged image files.

Indian Institute of Science Education and Research Kolkata (IISERK)

Summer Research Fellow, Advisor: Dr. Ayan Banerjee

May 2018 – Jul. 2018

Department of Physical Sciences

- Project Title: Use of optical tweezers as a measurement tool for precise diffusion coefficient determination
- Set up an optical experiment to trap and manipulate microbeads to study their kinematics.
- Calculated viscosity of fluids and compared the results with alternate techniques of analysis to study the precision of the method used.

University of Delhi (DU)

Summer Research Fellow, Advisors: Drs. Sourin Das and Ajit Mahapatro

May 2016 – Jul. 2016

Department of Physics and Astrophysics

- Project Title: UV Absorption and Photoluminescence Study of Alpha-Sexithiophene (6T)
- Learned about different techniques for making thin films, e.g. spin coating, physical vapor deposition.
- Prepared 6T thin film using the 'thermal deposition technique' and studied its optical properties with the help of spectrophotometers.

University of Delhi (DU)

Self-funded, Advisor: Dr. Pandian Senthil Kumar

May 2015 – Jul. 2015

Department of Physics and Astrophysics

- Project Title: The Science of Nanoparticles
- Synthesized and characterized gold and silver spherical nanoparticles and studied their size and surface area dependency for comparison and analysis.
- Explored their medicinal and industrial applications.

PUBLICATIONS

- A. Anker, [A. Avasthi](#), et al. **(2024)**, *arXiv, Xenon Detector $0\nu\beta\beta$ Searches: Steps Towards The Kilotonne Scale*.
doi: arXiv:2404.19050
- [A. Avasthi](#), T. Bezerra, et al. **(2022)**, *arXiv, Low Background kTon-Scale Liquid Argon Time Projection Chambers*.
doi: arXiv:2203.08821
- [A. Avasthi](#), TW Bowyer, et al. **(2021)**, *Physical Review D., Kilotonne-scale Xenon Detectors For $0\nu\beta\beta$ And Other New Physics Searches*.
doi: 10.1103/PhysRevD.104.112007

WORKSHOPS/CONFERENCES

- Giving an invited talk at the 'Summer Conference on High Energy Physics Astrophysics' on 'Liquid Xenon Time Projection Chamber for Rare Event Search.'
- Gave a talk during the mini-symposium on 'Particle Physics Experiments Using Noble Liquids' being at the APS Global Physics Meeting 2025, held in Anaheim, CA from March 16-21, 2025.
- Attending the APS NMC-Bridge Symposium, being held at 'The Joint March and April Meeting'.
- Presented at the university-level 'Three Minute Thesis (3MT) competition on February 28, 2024.
- Presented a poster on 'Novel Toxic Gases as Detector Media in TPCs' at the 'Coordinating Panel for Advanced Detectors (CPAD) Workshop 2024' at the University of Tennessee Knoxville, TN, jointly organized by the Oak Ridge National Lab from November 19-22, 2024.
- Participated in the 'Pitch Your Poster' competition held during the 'Innovation Week' at CWRU on November 1, 2024.
- Attended the 'Conference for Graduate Women and Gender Minorities in Physics 2024' held in Washington, DC from August 19-20, 2024.
- Attended the 'Cleveland Biomedical Trainee Alliance Conference' held at Case Western Reserve University, Ohio from December 7-8 2023.
- Attended the 'Workshop on Xenon Detector $0\nu\beta\beta$ Searches: Steps Towards the Kilo-tonne Scale' held at SLAC National Accelerator Laboratory, CA from October 25-27, 2023.

INTERNATIONAL INVOLVEMENT

- Served as a committee member of the 'International Student Advisory Council' of the International Student Services Center of CWRU to improve support for and engagement with the CWRU international student body.
- Represented Case Western Reserve University, USA at the international Tri-Institute TRISEP-2021 school on particle physics; collaborated with four other participants and presented ideas on a 'space-based futuristic neutrino detector.'
- Represented India at the international plasma physics summer school (PLAS@PAR) in Vietnam hosted by 'The International Center for Interdisciplinary Science and Education (ICISE)'; actively participated in lectures and group projects that involved remotely operated tokamak GOLEM, SMILEI PC simulations.
- Accepted into the annual Plasma Physics school organized in Paris, France; opted out due to conflicting commitments.

TECHNICAL MINI-PROJECTS

Micro-internship (Remote) Trade Study **Jul. 2024**

- SUPERVISOR: Mr. Steve Buscarello, President, Update International
- Assisted in developing a setup for measuring concentricity and selected the sub-components for a sensing axial run-out.

Triangle Wave Generator 741 IC hardware **Oct. 2018**

- SUPERVISOR: Dr. Alika Khare, IIT Guwahati
- Built an electronic circuit to demonstrate the working of a wave generator for sound synthesis and sampling circuits.

Heat Simulation Code C, Mathematica software **Jul. 2017**

- SUPERVISOR: Dr. Pankaj Kumar Mishra, IIT Guwahati
- Developed 150+ lines of code to implement the 'Crank-Nicholson Method' to simulate 1D and 2D heat transport in conductors.

Band Structure Calculations ABINIT software **Jan. 2017**

- SUPERVISOR: Dr. Kuldeep Kumar Kapil, SGTB Khalsa College

- Calculated the theoretical band structure diagram of graphite and correctly predicted its observed properties, demonstrating the theory's validity.

LCR Resonance Simulator *QUCS software*

Oct. 2016

- SUPERVISOR: Dr. Kuldeep Kumar Kapil, SGTB Khalsa College
- Demonstrated the analogy between an LCR circuit's resonant peak and a spherical nanoparticle's plasmonic frequency.

INSTRUCTIONAL EXPERIENCE AND ACADEMIC SERVICES

- Serving as teaching assistant for PHYS 302/318 'Advanced Physics Lab' and grader for PHYS 221 'Introduction to Modern Physics' in Spring 2025.
- Served as teaching assistant for PHYS 313 'Thermodynamics and Statistical Mechanics' and grader for PHYS 115 'Introductory Physics Lab' in Fall 2024.
- Served as a reviewer for the GCAS 'Graduate Student Scholarship and Creative Endeavors (GCSSE) 2022 grant review panel in the College of Arts and Sciences; selected 5 winning grants out of the 18 applicants for a reward of 1000\$ each.
- Submitted a proposal for 'The Graduate Student Scholarship and Creative Endeavors' grant hosted by the Graduate Council of Arts & Sciences, CWRU.
- Served as a live moderator, reviewer, and judge at 'Intersections: SOURCE symposium' for 10+ STEM seniors, presenting their senior thesis at Case Western Reserve University for Fall 2021 and Spring 2022.
- Taught 'Mechanics' and 'Electricity & Magnetism' undergraduate labs for a class of 70+ pre-med majors while working as a TA (online/hybrid mode), in addition to holding weekly office hours and serving as a grader for reports and final exams for over 200 students in Fall 2020 and Spring 2021.
- Taught 'Introductory Physics' lab to a class of 60+ physics majors while working as a TA (in-person mode), in addition to holding biweekly office hours, proctoring and working as a grader for reports and final exams of over 300 students in Fall 2019 and Spring 2020.
- Served as a mentor for incoming graduate students for 2021-23 as a part of the Physics Under-Represented Minority Students (PURMS) Group and high school students.

MAKERSPACE EQUIPMENT TRAININGS

- **Digital Prototyping:** Epilog Laser Cutter, Desktop FDM 3D Printer, Industrial Polyjet 3D Printer, Camerascope, Laser 3D Scanner
- **Non-digital Prototyping Tools:** Microscope for Viewing Circuit Boards, Vacuum Chamber, Vinyl Cutter, Soldering Iron, Desoldering Gun
- **Fabrication Tools:** CNC Lathe, CNC Mill, CNC Router, Waterjet Cutter, Drill Press, Grinder, Jointer, Planer, Panel Saw, Lathe, Metal Shop Sanders, MIG/TIG welder, Milter Saw, Spray-Paint, Sand Blaster, Vacuum Former, Vertical Bandsaw, Router Table, Wood Bandsaw

SKILLS

Programming Languages: C++, Python, HTML, Mathematica

Software Frameworks/Toolkits: ROOT, Geant4, Microsoft Office, Adobe, AutoDesk Fusion 360

Certifications: Powerful Presentation, Emotional Intelligence, Women in Leadership, Time Management Fundamentals

RELEVANT GRADUATE-LEVEL COURSEWORK

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|-------------------------|--------------------------------|---|
| • Mathematical Physics | • Electronics (Theory & Lab) | • Thin Film Phenomena |
| • Classical Mechanics | • Computational Physics | • Supervised Teaching |
| • Quantum Mechanics | • Atomic & Molecular Physics | • Elementary Particle Physics |
| • Computer Programming | • Nuclear & Particle Physics | • Research in Physics |
| • Statistical Mechanics | • Measurement Techniques | • Nuclear Physics & Technology |
| • Electrodynamics | • Advanced Physics Lab | • Frontiers in Physics |
| • Solid State Physics | • General Theory of Relativity | • Foundations of the Standard Model of Particle Physics |
| • Numerical Methods | • Elementary Particle Physics | |

AWARDS/HONORS

- Awarded the Distinguished Student Program (DS), for young researchers enrolled in the US, presenting at the APS Global Physics Summit 2025.
- Awarded the 'Most Intriguing Title' prize for a poster presentation on 'Novel Toxic Gases as Detector Media in TPCs' at the 'Coordinating Panel for Advanced Detectors (CPAD) Workshop 2024' at the University of Tennessee Knoxville, TN and the Oak Ridge National Lab.
- Awarded the 'Travel Funding Award for Early-Career Researchers' for the 'Coordinating Panel for Advanced Detectors (CPAD) Workshop 2024' held at the University of Tennessee, TN.
- Granted a merit-based scholarship of 7,50,000 INR (9036 USD) from the JN Tata Trust on January 23, 2024, India for excellent scholarly performance during PhD.
- Awarded the 'Student Leadership Award - Dignity of Difference' on April 20, 2023, in recognition of the excellence of the International Student Advisory Council.
- Presented with the 'Outstanding Student Organization' award by the Center for International Affairs on April 13, 2022, for the work done through the 'Graduate Council of Arts & Sciences.'
- Featured in the university's newsletter and social media and recognized by the Department of Arts and Sciences for my work as an advocate for the international student community of the university.
- Scored 56/60 on the SPEAK test (Speaking Proficiency English Assessment Kit) offered by the English Language Institute(ELI) as a part of the TA selection process for graduate school.
- Recipient of the prestigious 'JN Tata Fellowship Endowment for the Higher Education of Indians' in 2019 by The Tata Trust, India for pursuing doctoral studies abroad for five years; Included 100,000 INR as scholarship-cum-loan and 50,000 INR as travel grant in addition to gift scholarship based on performance during the doctorate degree.
- Scored 313/340 on the ETS-GRE with an 'Analytic Writing' score of 4.5/6 and the 82 percentile in 2018.
- Scored a total of 108/120 on the TOEFL-iBT test with 27 in reading, 30 in listening, 27 in speaking, and 24 in writing.
- Awarded the 'Life Skills' award for successfully completing tasks focused on developing soft skills within and outside the workplace in 2017.
- Cleared both the screening and the selection interview rounds after qualifying IIT-JAM to gain admission to the prestigious Indian Institute for Science, Education and Research (IISER) Pune for integrated M.Sc.-Ph.D. in Physics.
- Secured a place in the top 40 students of the country for admission to the prestigious Savitribai Phule Pune University.
- Secured an 'All India Rank' of 10 out of the 25,000 students appearing for the entrance examination for Masters in Physics at the University of Delhi.
- Placed among the national top 1% of the total enrollment of 14,000 students at the 'National Graduate Physics Examination' conducted by the Indian Association of Physics Teachers in 2017.
- Awarded the 'Sdn. Kauran Roop Rai Memorial-Best Female Student in Physics' scholarship and a cash prize based on performance in undergraduate courses in 2017.
- Selected three times for the all-India 'Summer Research Fellowship Program' consecutively for fully funded research in basic and natural sciences anywhere in India.
- Secured a place in the top 1% of the 1,290,028 applicants who appeared for the IIT-JEE exam, ranked the second toughest exam in the world.
- Awarded 400,000 INR (80,000 INR for 5 years) by the Department of Science and Technology-INSPIRE award of category SHE (Scholarship for Higher Education) from 2014-2019 for being in the top 1% of all qualified students in the CBSE board examinations.
- Qualified for the 'Gail Utkarsh Super-100' that covered full tuition and living expenses for the top 100 students ranked in their aptitude test taken by 2.2 million students.
- Awarded a certificate of merit in 2014 for securing a place within the top 0.1% of the 3.5 million students sitting for the English CBSE board examination.
- Awarded a merit certificate in 2013 for excellent performance in the following areas at the national level: Language Conventions: 98.0837%, Qualitative Reasoning: 95.8719%, Quantitative Reasoning: 95.4218%.
- Awarded the title of 'Amul Vidya Shree' by the 'Bharat Vikas Parishad' for being the topper of the batch of 2012 of 180 students in the high school examinations.
- Felicitated at the annual 'Guru Vandan Chhatra Abhinandan 2012' program organized by the Swami Vivekananda Foundation for securing the first rank in the cohort.

LEADERSHIP ROLES/EXTRACURRICULAR EXPERIENCES

- Served on the executive board as a treasurer of the 'Graduate Council of Arts and Sciences' for the academic year 2022-24; duties include administering the budget, maintaining financial records, identifying funding sources, applying for and managing funding from the graduate student governing body, post financial documents, disburse funds, and requests for up to 7,750\$.
- Served as the president and vice-president of the 'Local Inspirational Figures Talks (LIFT)' group for the academic year 2023-25 where we spread awareness of local inspirational figures and promote interdisciplinary collaboration and communication.
- Participated in the Mather Mentoring Program for the professional development of women; currently assigned as a mentee to the NASA aerospace engineer, Ms. Lizalyn Smith.
- Served as a SOURCE judge for undergraduates in physics and chemistry for three years consecutively.
- CPR-AED certified, 'Basic First Aid' and 'Stop The Bleed' trained.
- Temporary volunteer for the Cleveland Animal Protective League.
- Awarded the Federal Communications Commission (FCC) 'Amateur Extra' license after qualifying for the 'technician' and 'general' amateur radio licenses to serve as a ham radio emergency communicator for disaster management; qualified for the 2x2 call sign AD8IP.
- Completed laboratory training on Environmental, Health & Safety (EHS).
- Attended Mr. Paul Nowak's webinar on 'Speed Reading & Memory Techniques' in December 2019 as a part of the workplace efficiency promotion program.
- Participated in a week-long 'MATLAB workshop' organized by the IIT Guwahati Research Scholar Forum in 2019.
- Participated in a week-long 'L^AT_EXworkshop' organized by the IIT Guwahati Research Scholar Forum in 2019.
- Participated in the E-Commerce conclave organized in association with BRICS, Chamber of Commerce & Industry to provide a dynamic institutional connection for the promotion of trade and industries among the BRICS countries.
- Participated in the annual development fest for the social work society 'Leaders for Tomorrow' that included donation drives, cleanathons, volunteering to teach at local schools for the underprivileged, plantation programs, etc.
- Ranked in the top three for three consecutive years at the 'World Environment Day' themed poster and debate competitions organized annually by the Indian Institute of Toxicology Research.
- Taekwondo green-one belt holder.